Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning with research experience in Natural Language Processing, Large Language Models, Information Retrieval, and transformer-based learning. Experienced in designing reproducible experimental pipelines, evaluating deep learning models, and developing NLP applications using Python, PyTorch, Hugging Face, and FastAPI. Published IEEE researcher with a strong interest in Retrieval-Augmented Generation (RAG), multilingual NLP, explainable AI, and trustworthy AI systems.

SKILLS

- **Programming:** Python, Java, SQL, C++, Bash
- **Natural Language Processing & LLMs:** Hugging Face Transformers, Large Language Models (LLMs), Sentence Transformers, Retrieval-Augmented Generation (RAG), Transformer Models, Semantic Search, Information Retrieval
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Computer Vision, Vision Transformers, Continual Learning
- **Data Science & Analytics:** Pandas, NumPy, Statistical Analysis, Feature Engineering, Model Evaluation
- **Tools:** Git, Linux, Docker, AWS, FastAPI
- **Research Interests:** Large Language Models, Information Retrieval, Retrieval-Augmented Generation (RAG), Natural Language Processing, Explainable AI, Multilingual AI, Trustworthy AI

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PUBLICATION

Machine Learning for Air Pollution – IEEE Conference Publication (2023)

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

RESEARCH EXPERIENCE

Evaluation of Pretrained Vision Transformer Architectures Under Continual Learning Settings (Master's Thesis) **Nov 2025 - May 26**

- Conducted research on continual learning for Vision Transformer architectures, evaluating model robustness, catastrophic forgetting, and knowledge retention across sequential learning tasks.
- Reviewed and analyzed recent research literature on continual learning and transformer-based computer vision models to guide experimental design.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Evaluated model performance under changing data distributions and analyzed model reliability.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence application with backend APIs for automated document analysis and semantic comparison.
- Integrated Sentence Transformer models into inference pipelines for semantic similarity and document intelligence tasks.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Built semantic retrieval and document comparison pipelines using Sentence Transformers and embedding-based similarity for information retrieval and document intelligence.
- Leveraged transformer-based embeddings for semantic document understanding and retrieval-oriented workflows.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, Hugging Face Transformers, Sentence Transformers, FastAPI, Docker, AWS ECS, CloudWatch

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer-based embeddings, quantization techniques, and text reconstruction methods.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings